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△

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1 Introduction and overview

This protocol specification specifies the format, message sequences and semantics of the AUTOSAR Network Management (NM) protocol.

NM is intended to work together with an underlying communication stack, independent of the physical layer of the communication system used.

The AUTOSAR Network Management is a hardware independent protocol (for limitations refer to chapter 1.2.2).

The following figure shows how the NM interfaces with an upper (see 1.3.3) and a lower (bus) layer.

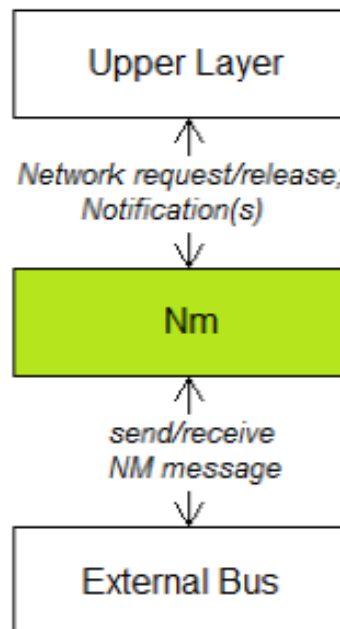


Figure 1.1: NM interfaces

1.1 Protocol purpose and objectives

Main purpose of the NM protocol is to coordinate one or more groups of ECUs to wake up and shutdown their communication stack synchronously.

The NM algorithm is based on periodic NM messages, which are received by all nodes in a NM cluster. Reception of NM messages indicates that sending nodes want to keep NM cluster awake. If any node does not need communication any more, it stops sending NM messages, but if NM messages from other nodes are received, it postpones transition to sleep mode. Finally, if a dedicated timer elapses because no NM messages are received anymore, every node initiates transition to the sleep mode, the NM node initiate the shutdown of the corresponding network.

If any node in the NM cluster requires bus-communication, it can keep the NM cluster awake by transmitting NM messages.

1.2 Applicability of the protocol

1.2.1 Constraints and assumptions

There may exist NM nodes in the network which follow the protocol without ever actively sending NM messages (called NM passive nodes).

1.2.2 Limitations

1. One NM instance is associated with only one NM cluster in one network. One NM cluster can have only one instance of Nm in one node.
2. The maximum size of the NM message is limited by the used communication bus.
3. The NM Coordinator is only used in the AUTOSAR Classic Platform. Details can be found in [1].

1.3 Dependencies

1.3.1 Dependencies to other protocol layers

NM algorithm uses services of the underlying communication stack modules to send and receive NM messages.

1.3.2 Dependencies to other standards and norms

N/A

1.3.3 Dependencies to the Application Layer

Upper layer (e.g. application) uses NM services to request or release a network i.e. to activate or deactivate sending of NM messages.

In addition, the upper layer/module may use the possibility to get informed about changes of the NM operational modes.

2 Use Cases

The Use Cases for PRS Network Management Protocol are listed and described in the chapter "Use Cases" of RS Adaptive Network Management [\[2\]](#).

3 Protocol Requirements

3.1 Requirements Traceability

Requirement	Description	Satisfied by
[RS_Nm_00044]	The Nm shall be applicable to different types of communication systems which are in the scope of AUTOSAR and support a bus sleep mode.	[PRS_Nm_00116] [PRS_Nm_00117] [PRS_Nm_00118] [PRS_Nm_00507]
[RS_Nm_00047]	Nm shall provide a service to request to keep the bus awake and a service to cancel this request.	[PRS_Nm_00005] [PRS_Nm_00237] [PRS_Nm_00334] [PRS_Nm_00337] [PRS_Nm_00503] [PRS_Nm_00504]
[RS_Nm_00048]	Nm shall put the communication controller into sleep mode if there is no bus communication	[PRS_Nm_00103] [PRS_Nm_00115] [PRS_Nm_00506]
[RS_Nm_00054]	There shall be a deterministic time from the point where all nodes agree to go to bus sleep to the point where bus is switched off.	[PRS_Nm_00103] [PRS_Nm_00115]
[RS_Nm_00150]	Specific features of the Network Management shall be configurable	[PRS_Nm_00013] [PRS_Nm_00045] [PRS_Nm_00074] [PRS_Nm_00075] [PRS_Nm_00158] [PRS_Nm_00328] [PRS_Nm_00335] [PRS_Nm_00338] [PRS_Nm_00339] [PRS_Nm_00405] [PRS_Nm_00406] [PRS_Nm_00407] [PRS_Nm_00408]
[RS_Nm_02503]	The Nm API shall optionally give the possibility to send user data	[PRS_Nm_00158]
[RS_Nm_02504]	The Nm API shall optionally give the possibility to get user data	[PRS_Nm_00158]
[RS_Nm_02505]	The Nm shall optionally set the local node identifier to the Nm-message	[PRS_Nm_00013] [PRS_Nm_00074]
[RS_Nm_02513]	Nm shall provide functionality which enables upper layers to control the sleep mode.	[PRS_Nm_00337] [PRS_Nm_00503]
[RS_Nm_02517]	CanNm shall support Partial Networking on CAN	[PRS_Nm_00328] [PRS_Nm_00332] [PRS_Nm_00333] [PRS_Nm_00341] [PRS_Nm_00412] [PRS_Nm_00413]
[RS_Nm_02519]	The Nm Control Bit Vector shall contain a PNI (Partial Network Information) bit.	[PRS_Nm_00328] [PRS_Nm_00329] [PRS_Nm_00331] [PRS_Nm_00335] [PRS_Nm_00340] [PRS_Nm_00409] [PRS_Nm_00410] [PRS_Nm_00411]
[RS_Nm_02541]	Nm shall define a common layout of Nm messages.	[PRS_Nm_00077] [PRS_Nm_00501] [PRS_Nm_00502] [PRS_Nm_00505]
[RS_Nm_02548]	<Bus>Nm PNC shutdown Propagation and Evaluation	[PRS_Nm_00406] [PRS_Nm_00409] [PRS_Nm_00411] [PRS_Nm_00412] [PRS_Nm_00413]
[RS_Nm_02561]	Nm shall support Partial Networking	[PRS_Nm_00338] [PRS_Nm_00339] [PRS_Nm_00407] [PRS_Nm_00408]

Table 3.1: Requirements Tracing

4 Definition of terms and acronyms

4.1 Acronyms and abbreviations

The glossary below includes acronyms and abbreviations relevant to the Network Management specification that are not included in the [3, AUTOSAR glossary].

Abbreviation / Acronym	Description
CBV	Control Bit Vector
FR	FlexRay
NM	Network Management
PN	Partial Network
PNC	Partial Network Cluster
PNI	Partial Network Information
PNL	Partial Network Learning

Table 4.1: Acronyms and abbreviations

4.2 Definition of terms

Term	Description
Network Mode	In this state the network is requested or active.
Prepare Bus-Sleep Mode	In this state the network is released or inactive.
Bus-Sleep Mode	In this state the network is released or inactive. In this state no NM message is sent
FlexRay communication cycle	Part of FlexRay communication schedule consisting of time slots (static or dynamic). Each FlexRay message is assigned to a specific time slot in one communication cycle.
NM cluster	Set of NM nodes coordinated with the use of the NM algorithm.
NM Message	Refers to the payload transmitted on the bus. It contains the User Data as well as the Control Bit Vector and may contain the Source Node Identifier.
Normal Operation	In this state the node is sending periodic NM messages in order to keep a NM cluster awake
Repeat Message State	This state ensures that transition, through a repetitive transmission of NM messages, to normal operation is visible for other nodes on the bus
Repeat Message Request	Request (received internally or externally via an NM message) to transition back to the Repeat Message State
NM Node	A ECU (electronic control unit) which is connected to one or more NM clusters
NM instance	A NM instance represents the current status of one NM cluster inside one NM node
External Request	Communication request via received NM message
Internal Request	Communication request via a NM node internal (request by application / upper layer)
Passive wakeup	A wakeup triggered by an external request
Active wakeup	A wakeup triggered by an internal request
PNC Bit Vector	Represent the Partial Network information in a NM frame
PNC Bit Vector Length	Represent the length of a Partial Network information in bytes

Term	Description
PNC bit	One bit with represent a particular Partial Network in the Partial Network Bit Vector Length
Top-level PNC coordinator	An ECU acts as top-level PNC coordinator for those PNCs which are actively coordinated on all assigned channels. This ECU has the PNC gateway functionality enabled. The top-level PNC coordinator triggers for those PNCs a synchronized PNC shutdown, if no other ECU in the network requests them and if the synchronized PNC shutdown is enabled. Note: For different PNCs it is possible to have different top-level PNC coordinators.
Intermediate PNC coordinator	An ECU acts as intermediate PNC coordinator for those PNCs which are passively coordinated on at least one channel. This ECU has the PNC gateway functionality enabled. The intermediate PNC coordinator forwards a synchronized PNC shutdown to active coordinated channels for PNCs which are passively coordinated, if the synchronized PNC shutdown is enabled.
PNC leaf node	A PNC leaf node is an ECU that act neither as top-level PNC coordinator nor as an intermediate PNC coordinator. It act as an ECU without a PNC gateway in the network and process PN shutdown message as usual NM messages.
PN shutdown message	A top-level PNC coordinator transmits the PN shutdown messages to indicate a synchronized PNC shutdown across the PN topology. A PN shutdown message is an NM message where the PNSR bit (resides in the control bit vector) and all PNC bits (reside in the PNC Bit Vector) which are indicated for a synchronized shutdown set to '1'. An intermediate PNC coordinator which receives a PN shutdown message forwards the PNC Bit Vector as a PN shutdown message on the affected channels. Note: An intermediate PNC coordinator has to forward the PNC Bit Vector of a received PN shutdown message as fast as possible to ensure a synchronized shutdown of the affected PNCs across the PN topology at nearly the same point in time.
PNC Gateway	A PNC Gateway is used to span (logical) partial network clusters across bus/communication channel boundaries, "gatewaying" PNC requests from one bus/network to the others. Therefore, a PNC gateway needs to be connected to multiple physical communication channels. The PNC gateway collects PNC requests from all of its multiple so-called "active" coordinated channels. The PNC gateway sends the aggregated PNC state in the network to all its active channels, which causes all nodes to have the same view on the global PNC request state as the gateway. If the PNC gateway is not the topmost PNC gateway in the network hierarchy, the PNC gateway will also send the aggregated PNC request state of all subordinate nodes, plus its own internal request state, to its superior PNC coordinator, which is connected via the so-called "passive" coordinated channel.
Active coordinated channel	A PNC gateway has communication channels which must be coordinated regarding the PNC requests. A PNC gateway collects PNC requests from all of its active coordinated channels, aggregates them and forwards it to all channels (independent if active or passive coordinated). Active coordinated channels are actively kept awake by this PNC gateway.

Term	Description
Passive coordinated channel	A PNC gateway has communication channels which must be coordinated regarding the PNC requests. A PNC gateway collects PNC requests from all of its passive coordinated channels, aggregates them and forwards it to all active coordinated channels. Passive coordinated channels are remotely kept awake by another PNC gateway, which is connected to the same channel and actively coordinates this channel.

Table 4.2: Definition of terms

5 Protocol specification

5.1 NM message format

[PRS_Nm_00501] Contents of an Nm Message

Status: DRAFT

Upstream requirements: [RS_Nm_02541](#)

[An Nm Message shall consist of the following elements:

- Control Bit Vector (CBV) of 1 Byte (optional)
- Source Node ID (SNI) of 1 Byte (optional)
- User Data of n Bytes (optional, may include PN Request Vector of n Bytes)

]

Note: There is an additional option to have a "Car Wakeup" bit.

[PRS_Nm_00502] Format of an Nm Message

Status: DRAFT

Upstream requirements: [RS_Nm_02541](#)

[User Data and/or PN Request Vector shall be located after CBV/SNI]

Note: UserData and PN Request Vector may overlap.

The following table shows an example layout of an NM message:

	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Byte 0	Control Bit Vector (default)							
Byte 1	Source Node Identifier (default)							
Byte 2	User data 0							
Byte 3	User data 1							
Byte 4	User data 2							
Byte 5	User data 3							
...	...							
Byte n	User data n-2							

Table 5.1: NM message layout example

[PRS_Nm_00505]

Upstream requirements: [RS_Nm_02541](#)

[The length of an NM message shall not exceed the MTU of the underlying physical transport layer.

]

[PRS_Nm_00077]

Upstream requirements: [RS_Nm_02541](#)

[The length (in bytes) of the NM message shall be configured by [\[NmMessageLength\]](#).]

Note: The length of the user data can be calculated from the [NmMessageLength](#) - (amount of used system bytes).

5.1.1 Source Node Identifier

[PRS_Nm_00074]

Upstream requirements: [RS_Nm_00150](#), [RS_Nm_02505](#)

[The location of the source node identifier shall be configurable to position Byte 0 or Byte 1 or Off (default: Byte 1). For FlexRay the source node identifier shall only be configurable to position Byte 1 or Off (default: Byte 1).]

[PRS_Nm_00013]

Upstream requirements: [RS_Nm_00150](#), [RS_Nm_02505](#)

[The source node identifier shall be available (set to a configurable value) unless the location of the source node identifier is set to Off.]

5.1.2 Control Bit Vector

[PRS_Nm_00075]

Upstream requirements: [RS_Nm_00150](#)

[The location of the Control Bit Vector shall be configurable to position Byte 0 or Byte 1 or Off (default: Byte 0). For FlexRay the Control Bit Vector shall be non-configurable and always be set to position Byte 0.]

[PRS_Nm_00045]

Upstream requirements: [RS_Nm_00150](#)

[The Control Bit Vector shall consist of:

- Bit 0: Repeat Message Request
 - 0: Repeat Message State not requested
 - 1: Repeat Message State requested
- Bit 1: PN Shutdown Request Bit (PNSR)

0: NM message does not contain synchronized Partial Network shutdown request

1: NM message does contain synchronized Partial Network shutdown request for at least one PNC

- Bit 3: NM Coordinator Sleep Ready Bit

0: Start of synchronized shutdown is not requested by main coordinator

1: Start of synchronized shutdown is requested by main coordinator

- Bit 4: Active Wakeup Bit

0: Node has not woken up the network (passive wakeup)

1: Node has woken up the network (active wakeup)

- Bit 5: Partial Network Learning Bit (PNL)

0: PNC learning is not requested

1: PNC learning is requested

- Bit 6: Partial Network Information Bit (PNI)

0: NM message contains no Partial Network request information

1: NM message contains Partial Network request information

- Bits 2,7 are reserved for future extensions

0: Disabled/Reserved for future usage

]

	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Byte 0/1	Reserved	Partial Network Information	Partial Network Learning	Active Wakeup	NM Co-ordinator Sleep Ready	Reserved	PN Shut-down Request Bit	Repeat Message Request

Table 5.2: CBV layout

Note: For FlexRay bit 7 is used as the Vote bit in certain schedule variants.

Note: Bit 1 and 2 were used in R3.2 as NM Coordinator ID (Low Bit)

5.1.3 User Data

User Data is considered all data not being part of CBV and NID.

[PRS_Nm_00158]

Upstream requirements: [RS_Nm_00150](#), [RS_Nm_02503](#), [RS_Nm_02504](#)

[It shall be possible to enable or disable the support of NM user data (NM user data is optional).]

5.2 Partial Networking

[PRS_Nm_00405]

Upstream requirements: [RS_Nm_00150](#)

[It shall be possible to enable or disable the PN support (PN feature is optional).]

[PRS_Nm_00406]

Upstream requirements: [RS_Nm_00150](#), [RS_Nm_02548](#)

[It shall be possible to enable or disable the handling of synchronized PNC shutdown (handling is optional). If handling is enabled, then also PN support shall be enabled.]

[PRS_Nm_00335]

Upstream requirements: [RS_Nm_00150](#), [RS_Nm_02519](#)

[If PN Support is enabled, the layout of the PNC Bit Vector within the NM message shall be pre-configured with [PnInfoOffset](#) and [PnInfoLength](#) (in bytes).]

Note:

Every bit (PNC bit) of the PNC Bit Vector Length represents one Partial Network. The following interpretation has to be considered:

- PNI bit = "1" and PNSR = "0": If the PNC bit is set to 1 the Partial Network is requested. If the bit is set to 0 there is no request for this PN.
- PNI bit = "1" and PNSR = "1" (received by a top-level PNC coordinator): ignore the PNSR bit and handle the message as a usual NM message. A top-level PNC coordinator should never receive a PN shutdown request. This is an error case, where an intermediate PNC coordinator or PNC leaf node sets the PNSR bit within the Nm message by accident.
- PNI bit = "1" and PNSR = "1" (received by an intermediate PNC coordinator): All the Partial Network were the corresponding PNC bits in the PNC Bit Vector are set to 1 are indicated to be released. The remaining Partial Network (the corresponding PNC bits are set to 0) are not affected.
- PNI bit = "1" and PNSR = "1" (received by a PNC leaf node): same as if PNI bit = "1" and PNSR = "0".

Note: Each bit (PNC bit) of the PNC Bit Vector represent a particular PNC. The [byteIndex](#) and [bitIndex](#) within the PNC Bit Vector of a PNC bit can be determined as

follows: $\text{byteIndex} = (\text{PncId} \div 8)$; $\text{bitIndex} = (\text{PncId} \bmod 8)$.

[PRS_Nm_00338]

Upstream requirements: [RS_Nm_00150](#), [RS_Nm_02561](#)

[If the PN Support is enabled, and if a message containing a PNC bit set to 1 is received, and the ECU is interested in this PNC, that PNC shall be considered "externally requested".]

[PRS_Nm_00407]

Upstream requirements: [RS_Nm_00150](#), [RS_Nm_02561](#)

[If the PN Support is enabled, and if a message containing a PNC bit set to 0 is received, and the ECU is interested in this PNC, that PNC shall be considered "externally released".]

[PRS_Nm_00339]

Upstream requirements: [RS_Nm_00150](#), [RS_Nm_02561](#)

[If the PN Support is enabled, and if one or more applications are requesting a PNC, and the ECU is interested in this PNC, this PNC shall be considered "internally requested".]

[PRS_Nm_00408]

Upstream requirements: [RS_Nm_00150](#), [RS_Nm_02561](#)

[If the PN Support is enabled, and if no application of an ECU is requesting a PNC anymore, then this PNC shall be considered as "internally released".]

5.2.1 Handling of Rx NM messages

[PRS_Nm_00328]

Upstream requirements: [RS_Nm_00150](#), [RS_Nm_02517](#), [RS_Nm_02519](#)

[If PN support is disabled, then its NM shall ignore any partial networking information contained in the received message.]

[PRS_Nm_00329]

Upstream requirements: [RS_Nm_02519](#)

[If the PN support is enabled, and the PNI bit in the received NM message is 0, the node's NM shall ignore the partial networking information bytes of the message.]

[PRS_Nm_00331]

Upstream requirements: [RS_Nm_02519](#)

[If the PN support is enabled, the PNI bit is set to 1 and the PNSR bit is set to 0 in the received NM message, NM shall process the Partial Networking Information of the NM message.]

[PRS_Nm_00409]

Upstream requirements: [RS_Nm_02519](#), [RS_Nm_02548](#)

[If synchronized PNC shutdown is enabled, a NM message is received in the role of a top-level PNC/intermediate PNC coordinator on an active coordinated channel and PNI bit and PNSR bit are set to 1, then NM shall ignore the PNSR bit and handle the message as a usual NM message.]

Note: A PN shutdown message (PNI bit = 1 and PNSR bit = 1) should never be received by a top-level/intermediate PNC coordinator on an active coordinated channel, because only a top-level PNC coordinator of a dedicated PNC could initiate a PN shutdown message. This is an error case where an intermediate PNC coordinator transmits a PN shutdown message by accident on an active coordinated channel. A receiving top-level/intermediate PNC coordinator should handle this message as a usual NM message.

[PRS_Nm_00410]

Upstream requirements: [RS_Nm_02519](#)

[If the PN synchronized shutdown error reaction is enabled and the received NM message is discarded due to [\[PRS_Nm_00409\]](#), then the top-level PNC coordinator shall immediately transmit an NM message with all "internally requested" and "externally requested" PNCs as Partial Network Information.]

[PRS_Nm_00411]

Upstream requirements: [RS_Nm_02519](#), [RS_Nm_02548](#)

[If synchronized PNC shutdown is enabled, an NM message is received in the role of an intermediate PNC coordinator on a passive coordinated channel and PNI bit and PNSR bit are set to 1, then NM shall release the indicated PNCs (PNC bits which are set to 1 within the PNC bit vector), reset the PN reset timer and forward the received NM message with PNI bit and PNSR bit set to 1 and the according PNCs set to 1 to all subordinated ECUs.]

Note:

- An intermediate PNC coordinator has to forward the received NM message to all remaining communication channels.
- Subordinated ECUs could be either further intermediate PNC coordinators and/or PNC leaf nodes.

- A PNC leaf node has no special handling upon reception of a PN shutdown message. It just handles the received NM message as specified in [PRS_Nm_00331].

[PRS_Nm_00340]

Upstream requirements: [RS_Nm_02519](#)

[If the PN support is enabled, and if one PNC is not requested again (relevant PNC bit is not set to 1 again) within [PnResetTime] this PN shall be considered as "not requested".]

Note: [PnResetTime](#) is configured to a value greater than [NmMsgCycleTime](#).

5.2.2 Handling of Tx NM messages

[PRS_Nm_00332]

Upstream requirements: [RS_Nm_02517](#)

[If the PN support is enabled, its NM shall set the value of the transmitted PNI bit in the CBV to 1.]

[PRS_Nm_00333]

Upstream requirements: [RS_Nm_02517](#)

[If the PN support is disabled, its NM shall set the value of the transmitted PNI bit in the CBV to 0.]

[PRS_Nm_00341]

Upstream requirements: [RS_Nm_02517](#)

[If the PN support is enabled, for PNCs that are "internally requested" the corresponding bit in the PNC Request Bit Vector shall be set to 1 before sending the NM message.]

Constraint: The usage of the CBV is mandatory in case Partial Networking is used. This must be ensured by configuration in the respective platform.

[PRS_Nm_00412]

Upstream requirements: [RS_Nm_02517](#), [RS_Nm_02548](#)

[If the PN support is enabled, for PNCs that are "internally requested" or "externally requested" the corresponding bit in the PNC Request Bit Vector shall be set to 1 before sending the NM message in the role of a top-level PNC coordinator or an intermediate PNC coordinator.]

[PRS_Nm_00413]

Upstream requirements: [RS_Nm_02517](#), [RS_Nm_02548](#)

[If synchronized PNC shutdown is enabled and NM detect an transition of PNCs from "requested" to "released" (independent if externally or internally requested), the cor-

responding bit of those released PNCs shall be set to 1, the remaining shall be set 0 and the PNSR bit in CBV shall be set to 1 before sending the PN shutdown message.]

5.3 Timing behavior

5.3.1 Transition to Bus-Sleep Mode

When a NM node does not need the communication on a bus, it will not immediately shut down i.e. switch to Bus-Sleep Mode. Instead, it will first change to the so called Ready Sleep state. This state ensures that any NM node in the NM cluster waits to transition to the Bus-Sleep Mode as long as any other node keeps the NM cluster awake.

[PRS_Nm_00103]

Upstream requirements: [RS_Nm_00048](#), [RS_Nm_00054](#)

[If bus communication is released, the NM algorithm shall perform transition to the Bus-Sleep Mode after a configurable amount of Ready Sleep Time has expired and no new communication request occurs in between and no NM Message has been received.]

Note: The Ready Sleep Time depends on the used network, refer to [5.4](#).

5.4 Networks specifics

5.4.1 CAN and Ethernet

On the transition path from Network to Bus-Sleep Mode, CAN NM and UDP NM introduce Prepare Bus Sleep Mode. The purpose of this state is to ensure that all nodes have time to stop their network activity before the Bus Sleep state is entered.

[PRS_Nm_00506]

Status: DRAFT

Upstream requirements: [RS_Nm_00048](#)

[If [NmStayInPbsEnabled](#) is disabled and bus communication in a NM cluster is released, and there are no Network Management PDUs on the bus for a configurable amount of time determined by NM Timeout Time ([NmTimeOutTime](#)) + Wait Bus-Sleep Time ([NmWaitBusSleepTime](#)) transition into the BusSleep Mode shall be performed.]

[PRS_Nm_00115]

Upstream requirements: [RS_Nm_00048](#), [RS_Nm_00054](#)

[If [NmStayInPbsEnabled](#) is disabled the NM shall stay in the Prepare Bus-Sleep Mode for an amount of time determined by the Wait Bus-Sleep Time. After this time

has expired, the Prepare Bus-Sleep Mode shall be left, and the Bus-Sleep Mode shall be entered.]

Note: Thus the Ready Sleep Time is extended by Wait Bus-Sleep Time ([NmWaitBusSleepTime](#)). The Ready Sleep Time on CAN and Ethernet starts when bus communication is released and it ends NM Timeout Time ([NmTimeOutTime](#)) after last NM messages was transmitted or received on the bus.

[PRS_Nm_00504]

Status: DRAFT

Upstream requirements: [RS_Nm_00047](#)

[When in Prepare Bus-Sleep Mode, and an NM message is received, or the NM is requested for communication, than NM shall enter Network Mode.]

[PRS_Nm_00507] Handling multiple network requests

Status: DRAFT

Upstream requirements: [RS_Nm_00044](#)

[If in Ready Sleep State, Normal Operation State or Repeat Message State and ([NmPnHandleMultipleNetworkRequests](#)) is set to true and the requested state of the channel/VLAN or an associated PNC changes, Repeat Message State shall be (re)entered.]

The following requirements concerns early shutdown. It does only apply to CAN and Ethernet, since on FlexRay always all messages have to be considered (FlexRay cannot shutdown earlier nodes):

[PRS_Nm_00503]

Status: DRAFT

Upstream requirements: [RS_Nm_00047](#), [RS_Nm_02513](#)

[It shall be possible to enable or disable All Nm Messages Keep Awake functionality.]

[PRS_Nm_00337]

Upstream requirements: [RS_Nm_00047](#), [RS_Nm_02513](#)

[If PN support is enabled, and All Nm Messages Keep Awake functionality is disabled, the NM algorithm shall only process messages if they contain at least one bit set to 1 in the PNC Request Bit Vector, that corresponds to a PNC which is relevant for the ECU.]

5.4.1.1 Sending NM message

If communication on the bus is needed i.e. requested, NM messages are sent out. If no communication is needed i.e. released, sending of NM messages is stopped.

[PRS_Nm_00237]

Upstream requirements: [RS_Nm_00047](#)

[NM messages shall be sent periodically in states "Repeat Message" and "Normal Operation" using configured NM Message Cycle Time ([NmMsgCycleTime](#)).]

[PRS_Nm_00005]

Upstream requirements: [RS_Nm_00047](#)

[If the "Repeat Message" state is not entered because of network request OR NM transmissions is zero (see [NmImmediateNmTransmissions](#)) the transmission of NM messages shall be delayed by NM Cycle Offset (see [NmMsgCycleOffset](#)) after entering the "Repeat Message" state.]

[PRS_Nm_00334]

Upstream requirements: [RS_Nm_00047](#)

[When the "Repeat Message" state is entered because of network request or repeat message request and configured number of immediate NM transmissions is greater than zero (see [NmImmediateNmTransmissions](#)), these immediate NM messages shall be transmitted using Immediate NM Cycle Time (see [NmImmediateNmCycleTime](#)). NM Cycle Offset (see [NmMsgCycleOffset](#)) shall not be applied in this case.]

5.4.2 FlexRay

In addition to NM message containing data (see Figure 5.1), the FlexRay NM specifies so-called NM-Vote messages.

In fact, the FlexRay NM algorithm is based on periodic NM-Vote messages received by all nodes in the cluster. Reception of a NM-Vote message indicates that the sending node wants to keep the NM cluster awake.

	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Byte 0	Vote	Set to "0"						

Table 5.3: NM-Vote message layout

[PRS_Nm_00116]

Upstream requirements: [RS_Nm_00044](#)

[The NM-Vote message format shall contain a Voting Bit (Vote) with the following meaning:

- 0 - vote against keeping awake
- 1 - vote for keeping awake

]

[PRS_Nm_00117]

Upstream requirements: [RS_Nm_00044](#)

[The FlexRay NM shall be able to separately transmit NM-Data and NM-Vote, or to combine them within one NM message (in either static or dynamic slot). Transmission format shall be configurable (Schedule Variant).]

When the NM-Vote and NM-Data are combined (by Bit OR-ing) within one NM message, the content of the NM-Vote will be combined with the content of the Control Bit Vector Byte.

	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Byte 0	Vote	Partial Network Information	Partial Network Learning	Active Wakeup	NM Co-ordinator Sleep Ready	Reserved	Reserved	Repeat Message Request

Table 5.4: Combined NM-Vote and CBV

Each ECU, which participates in the FlexRay NM, is synchronized to a global time based on periodic repetition of the FlexRay communication cycle. To assure synchronous behaviour of all ECUs in a NM cluster, the FlexRay NM aligns the state changes to a NM Repetition Cycle, which is aligned to a FlexRay communication cycle.

Every transition is bound to repetition cycles (refer to configuration parameter `NmRepetitionCycle`). Therefore the Ready Sleep Time is defined as the time that starts when a new repetition cycle starts after bus communication has been released and ends `NmReadySleepCnt+1` repetition cycles without any NM-Vote.

[PRS_Nm_00118]

Upstream requirements: [RS_Nm_00044](#)

[The FlexRay NM shall specify the following cycle configuration parameters:

- Voting Cycle - number of cycles needed to transmit NM-Vote of every node at least once
- Data Cycle - number of cycles needed to transmit the NM-Data of every node at least once
- Repetition Cycle - number of repetitions of Voting Cycle

]

Note: Further details can be found in the AUTOSAR SWS FlexRay specifications.

5.5 Sequences

5.5.1 Communication request

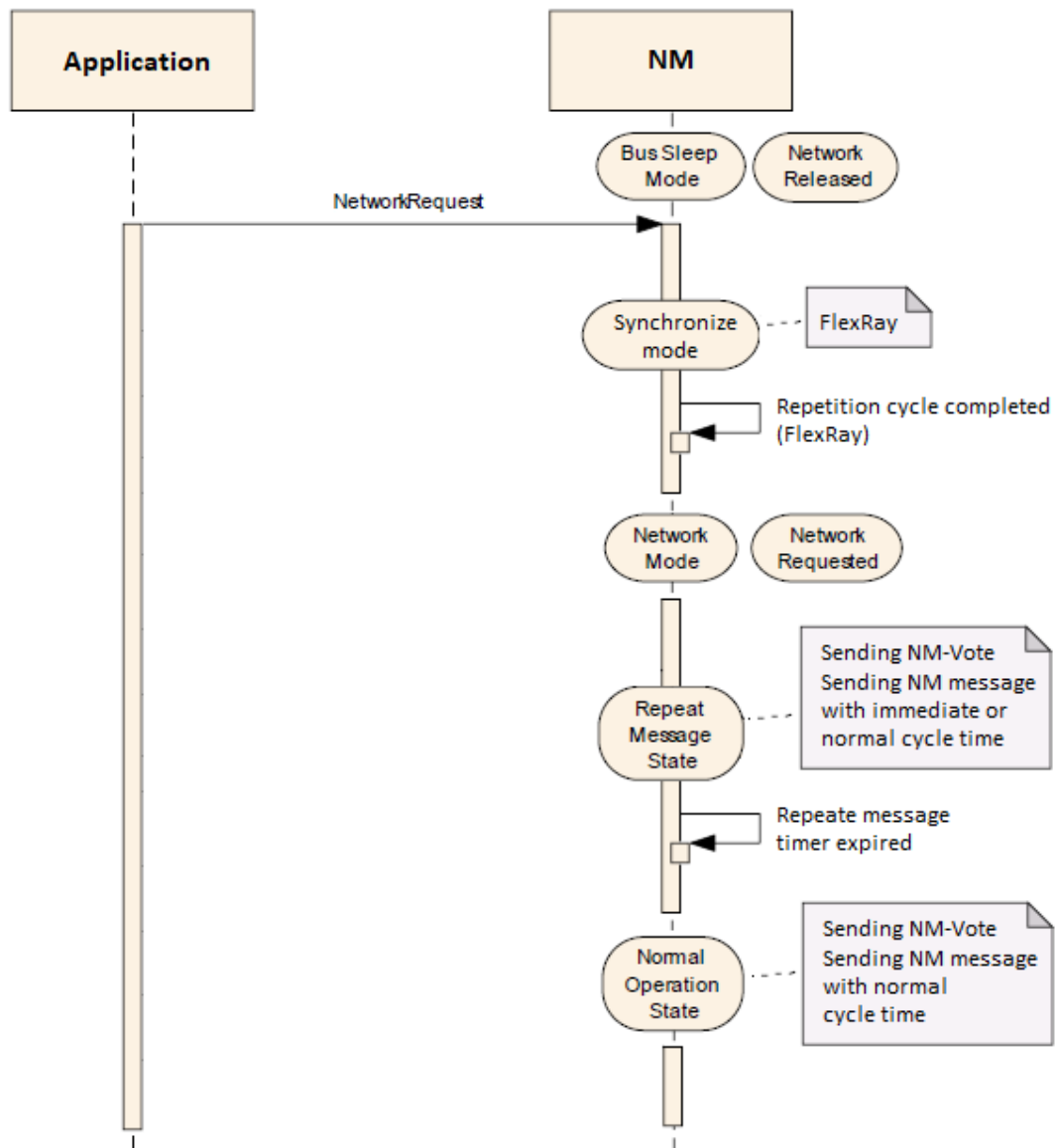


Figure 5.1: Communication request

5.5.2 Communication release

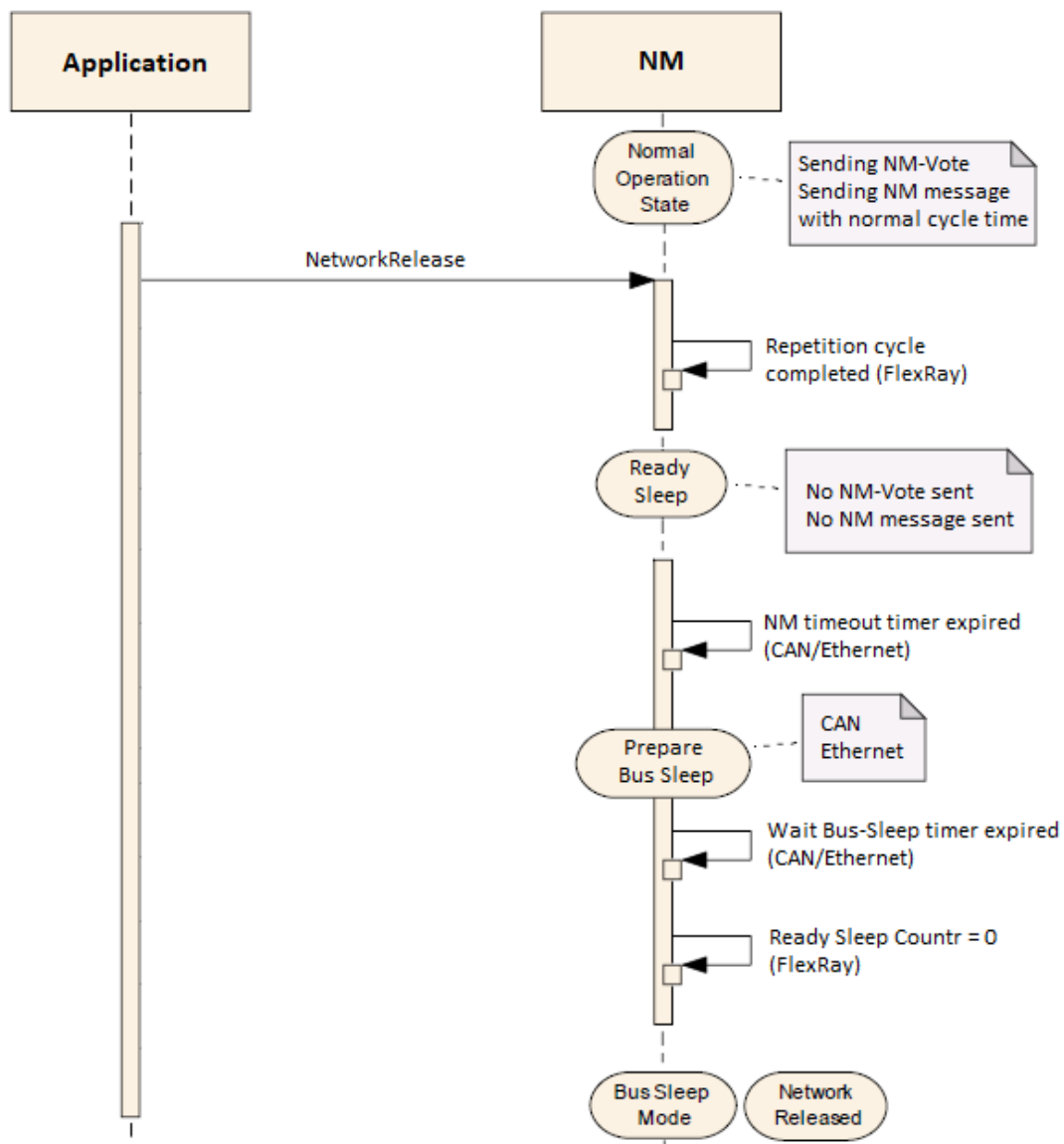


Figure 5.2: Communication release

6 Configuration parameters

This chapter lists all parameters the NM protocol uses.

6.1 NM Message Layout

Parameter	Description
NmNidPosition	Defines the position of the source node identifier (if used) within the NM message
NodeId	Node identifier of local node
NmCbvPosition	Defines the position of the Control Bit Vector (if used) within the NM message
UserDataEnabled	Enables/disables user data support
NmMessageLength	Specifies the length (in bytes) of the NM message
PnEnabled	Enables/disables support of partial networking
PnInfoOffset	Offset of the PN request information in the NM message. The offset to the PNC data can only be configured to either be directly after the system bytes or after the user data.
PnInfoLength	Length of the PN request information in the NM message

Table 6.1: NM Message Layout

6.2 Timeout Parameters

Parameter	Description
NmTimeOutTime	The time for a node between the reception of the last NM message keeping it awake to the transition to Bus Sleep
NmMsgCycleTime	The transmission periodicity of an NM message by a node
NmMsgCycleOffset	The start delay of a transmission of Nm messages in Repeat Message State
NmRepeatMessageTime	The time for a node to remain in Repeat Message State
NmWaitBusSleepTime	Timeout for bus calm down phase. It denotes the time in seconds how long the NM shall stay in the Prepare Bus-Sleep Mode before transition into Bus-Sleep Mode (CAN NM, UDP NM only).
NmReadySleepCnt	Ready sleep counter. After NmReadySleepCnt+1 repetition cycles without any NM-Vote, NM enters Bus-Sleep (FR NM only).
NmImmediateNmCycleTime	Defines the immediate NM message cycle time in seconds used in Repeat Message state (CAN NM, UDP NM only)
NmImmediateNmTransmissions	Number of immediate NM messages which shall be transmitted in Repeat Message state (CAN NM, UDP NM only)
NmDataCycle	Number of FlexRay Schedule Cycles needed to transmit NM-Data of all ECUs (FR NM only)
NmVotingCycle	Number of FlexRay Schedule Cycles needed to transmit NM-Vote of all ECUs (FR NM only)
NmRepetitionCycle	Number of NM voting cycles where no change of voting behavior is possible (FR NM only)
NmScheduleVariant	Defines the transmission scheduling variant for sending NM-Vote and NM-Data (FR NM only)

Parameter	Description
PnResetTime	Time a PNC is considered requested externally after the last message containing the corresponding bit set to one has been received

Table 6.2: Timeout Parameters

6.3 NM local configuration

Parameter	Description
SynchronizedPncShutdownEnabled	Enable/Disable a synchronized PNC shutdown
NmStayInPbsEnabled	If this parameter is disabled Prepare Bus-Sleep Mode is left after NmWaitBusSleep Time. If this parameter is enabled Prepare Bus-Sleep Mode can only be left if ECU is powered off or any restart reason applies.
NmPnHandleMultipleNetworkRequests	If enabled, an additional transition from Network Mode to Repeat Message State shall be performed, if the requested state of the channel/VLAN or an associated PNC changes.

Table 6.3: NM local configuration

7 Protocol usage and guidelines

No additional guidelines or How-To instructions for implementer.
All relevant information already provided in previous chapters.

8 References

- [1] Specification of Network Management Interface
AUTOSAR_CP_SWS_NetworkManagementInterface
- [2] Requirements on AUTOSAR Network Management
AUTOSAR_FO_RS_NetworkManagement
- [3] Glossary
AUTOSAR_FO_TR_Glossary

A Change history of AUTOSAR traceable items

Please note that the lists in this chapter also include traceable items that have been removed from the specification in a later version. These items do not appear as hyperlinks in the document.

A.1 Traceable item history of this document according to AUTOSAR Release R25-11

A.1.1 Added Specification Items in R25-11

Number	Heading
[PRS_Nm_00507]	Handling multiple network requests

Table A.1: Added Specification Items in R25-11

A.1.2 Changed Specification Items in R25-11

none

A.1.3 Deleted Specification Items in R25-11

none

A.2 Traceable item history of this document according to AUTOSAR Release R24-11

A.2.1 Added Specification Items in R24-11

none

A.2.2 Changed Specification Items in R24-11

none

A.2.3 Deleted Specification Items in R24-11

none

A.3 Traceable item history of this document according to AUTOSAR Release R23-11

A.3.1 Added Specification Items in R23-11

Number	Heading
[PRS_Nm_00005]	
[PRS_Nm_00505]	
[PRS_Nm_00506]	

Table A.2: Added Specification Items in R23-11

A.3.2 Changed Specification Items in R23-11

Number	Heading
[PRS_Nm_00115]	
[PRS_Nm_00334]	

Table A.3: Changed Specification Items in R23-11

A.3.3 Deleted Specification Items in R23-11

none

A.4 Traceable item history of this document according to AUTOSAR Release R22-11

A.4.1 Added Specification Items in R22-11

none

A.4.2 Changed Specification Items in R22-11

Number	Heading
[PRS_Nm_00335]	
[PRS_Nm_00337]	
[PRS_Nm_00341]	
[PRS_Nm_00406]	



△

Number	Heading
[PRS_Nm_00407]	
[PRS_Nm_00408]	
[PRS_Nm_00409]	
[PRS_Nm_00410]	
[PRS_Nm_00411]	
[PRS_Nm_00412]	
[PRS_Nm_00413]	

Table A.4: Changed Specification Items in R22-11

A.4.3 Deleted Specification Items in R22-11

Number	Heading
[PRS_Nm_00102]	

Table A.5: Deleted Specification Items in R22-11